**Ampere's Law:**

By this law of the magnitude i.e., the magnetic field intensity produced due to the current flow through a conductor can be determined. It establishes the relation between the current $i$ flowing through that conductor and the magnetic field $\vec{B}$ produced in it. The law is as follows:

"*The line integral of a magnetic field along a close path is equal to $\mu_0$ times, the net current flowing through the area subtended by the path*"

That means –

$$\oint \vec{\boldsymbol{B}}.\overrightarrow{\boldsymbol{dl}} = \boldsymbol{\mu_0 i}$$

Here, $\mu_0 =$ magnetic permeability of vacuum, $dl =$ displacement vector of path difference, $\oint =$ integration in a close path.

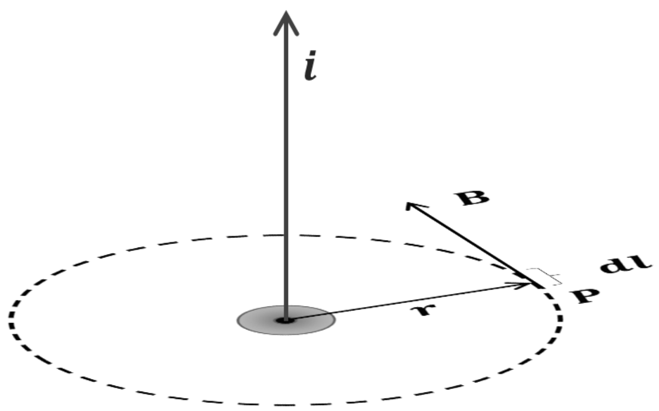
If the closed path does not link with the current loop, then –

$$\oint \vec{\boldsymbol{B}}.\overrightarrow{\boldsymbol{dl}} = \mathbf{0}$$

**Applications of Ampere's Law:** It can be applied to determine the magnetic field $B$ –

**a) For the long straight current carrying wire:**

Let us consider a long straight wire is carrying current $i$. We want to find the magnetic field at a point $P$ a distance $r$ from the wire. Figure shows the geometry for the calculation.




The lines of magnetic induction for a long straight wire carrying a current $i$ are concentric circles centered on the wire. Hence the point $P$ may be regarded to be on a circular loop of the radius $r$ surrounding the wire. Applying Ampere's law, we have –

$$\oint \vec{B}.\vec{dl} = \mu_0 i$$

$$<=> \oint Bdlcos\theta = \mu_0 i$$

Where, $\theta$ is the angle between the directions of $B$ and $dl$. Now $B$ has the same magnitude on any point on the circle. Moreover, $B$ and $dl$ point in the same direction, i.e., $\theta = 0^0$, then –

$$B \oint dl = \mu_0 i$$

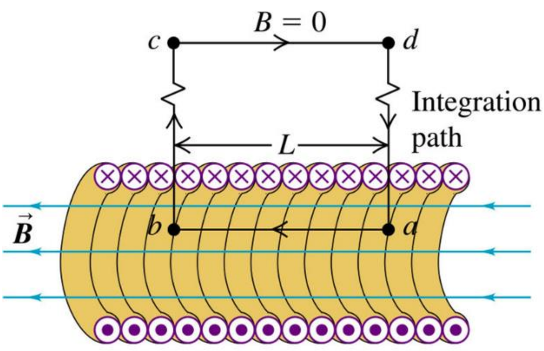
$$\Leftrightarrow \quad B.2\pi r = \mu_0 i$$

$$\therefore \quad \boldsymbol{B = \frac{\mu_0 i}{2\pi r}}$$

This equation is similar to that obtained by Biot-Savart law for long straight wire.

**b) For a long Solenoid:**

Figure shows the section of the solenoid, the dots indicating that current in the wire is outward from and the crosses inward to the page.



Central part of solenoid

Application of Ampere's law to the rectangular path $abcd$ in the direction shown in the figure gives –

$$\oint B.dl = \oint_a^b B.dl + \oint_b^c B.dl + \oint_c^d B.dl + \oint_d^a B.dl$$

Now, since $B$ is perpendicular to the sides $bc$ and $da$ the second and fourth integrals on the left side do not contribute. Also, $B$ vanishes outside the solenoid. Hence the third integral also contributes nothing. The net contribution comes only from the first integral. For this path, $B$ is parallel to $dl$ and constant at all points. Thus –

$$\oint B.dl = \oint_a^b B.dl = B\oint_a^b dl = BL$$

Where, $L$ is the length of the path $ab$. The net current $I$ that passes through the area bounded by the path $abcda$ is not the same as the current $I_0$ in the solenoid. Let $n$ be the number of turns per unit length of the solenoid. Then, $I = I_0 nL$

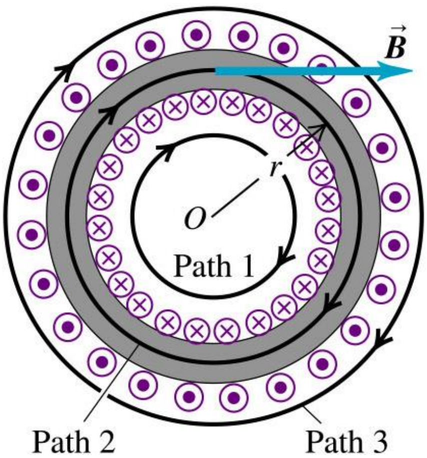
Ampere's law then becomes –

$$BL = \mu_0 I = \mu_0 I_0 nL$$

$$\boldsymbol{B = \mu_0 I_0 n}$$

**b) For a Toroid:**

Figure shows the cross-section of a toroid. Again dots and crosses denoted the direction of the current in the wire of the toroid. From symmetry, the lines of $B$ from concentric circles inside the toroid, as shown in the figure.




Application of Ampere's law to the circular path of the radius $r$, gives –

$$B\oint dl = \mu_0 I$$

$$\Leftrightarrow \quad B.2\pi r = \mu_0 I$$

$$\therefore \quad B = \frac{\mu_0 I}{2\pi r}$$

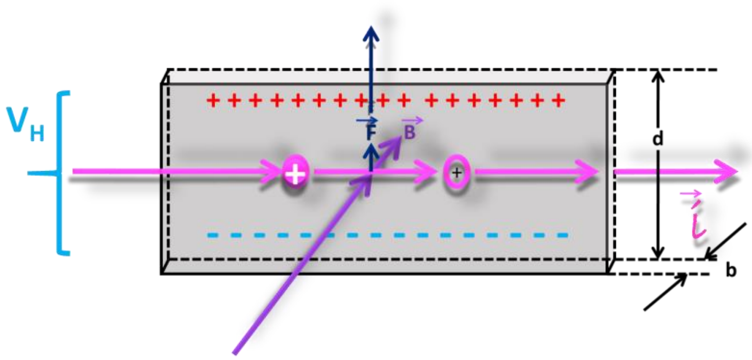
If $I_0$ is the current in the toroid wire and $N$ is the total number of the windings, then –

$$\boldsymbol{B = \frac{\mu_0 N I_0}{2\pi r}}$$

**Hall Effect:**

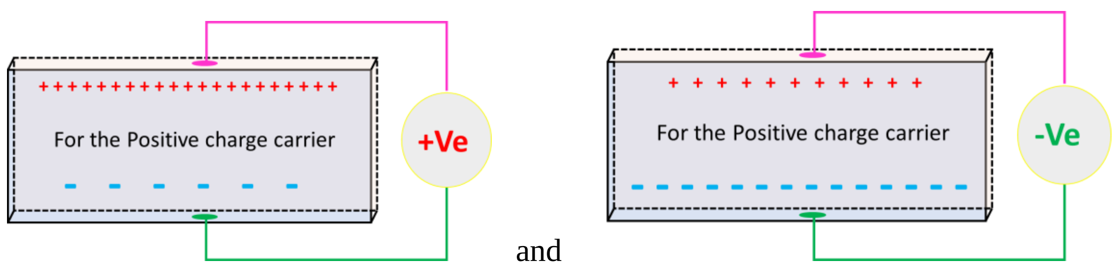
The Hall Effect is the production of a voltage difference (the Hall voltage) across an electrical conductor, transverse to an electric current in the conductor and to an applied magnetic field perpendicular to the current. It was discovered by Edwin Hall in 1879.

If an electric current flows through a conductor in a magnetic field, the magnetic field exerts a transverse force on the moving charge carriers which tends to push them to one side of the conductor. This is most evident in a thin flat conductor as illustrated. A buildup of charge at the sides of the conductors will balance this magnetic influence, producing a measurable voltage between the two sides of the conductor. The presence of this measurable transverse voltage is called the Hall Voltage.




**Applications of Hall Effects:** From the Hall voltage, it can be determined that –

(a) ***The nature of the charges:*** (for detail follow the recommended books)




***(b) The number of charges per unit volume:*** (for detail follow the recommended books)

The equation of the Hall voltage:

$$V_H = \frac{BI}{ntq}$$

***(Some recommended mathematical problem should be solved)***

***Problem 1:*** A solenoid is $1.0m$ long and $3.0cm$ in mean diameter. It has five layers of windings of 850 turns each and carries current of $5.0amp$. (a) What is $B$ at its centre? (b) What is the magnetic flux $\varphi_B$ for a cross-section of the solenoid at this centre?

***Problem 2:*** An air-core solenoid with 2000 loops on it is $600mm$ long and has diameter of $20mm$. If a current of $5A$ is sent through it, what will be the flux density within?

***Problem 3:*** A solenoid $0.5m$ long has 2000 turns. The magnetic induction near the centre of the solenoid is $0.08T$. What is the current in the solenoid?